

Manual Jenkins Installation

Kubernetes kubeadm home lab | Step-by-step reference | October 2026

Cluster: master 192.168.178.49; node01 192.168.178.51; node02 192.168.178.52. Kubernetes v1.28.15, Rocky Linux 9.7, containerd. Jenkins controller and storage run on node01; agents can be configured on node02 later.

1. Architecture and prerequisites

We use a dedicated namespace, a static local PersistentVolume (PV), a matching PersistentVolumeClaim (PVC), namespace-scoped ServiceAccount/RBAC, a StatefulSet, a headless Service for StatefulSet identity, and a NodePort Service for LAN access.

```
kubectl get nodes -o wide
kubectl get pods -n kube-system
kubectl get storageclass
kubectl get pv
kubectl get pvc -A
```

Note: All nodes should be Ready. kubeadm does not install a dynamic storage provisioner by default. Kubernetes 1.28 is outside upstream support; plan an upgrade separately.

2. Prepare storage on node01

Log in to node01 and create the Jenkins home directory. UID/GID 1000 match the standard Jenkins container user.

```
ssh root@192.168.178.51
mkdir -p /data/jenkins
chown 1000:1000 /data/jenkins
chmod 750 /data/jenkins
ls -ld /data/jenkins
getenforce
exit
```

Note: SELinux Permissive does not enforce denials, but standard Linux permissions still apply. Local PV data is tied to node01; 10Gi is an advertised PV size, not a filesystem quota.

3. Namespace and manifest workspace

```
mkdir -p /root/jenkins && cd /root/jenkins
kubectl create namespace jenkins
```

4. Static PersistentVolume

```
cat > jenkins-pv.yaml <<'EOF'
apiVersion: v1
kind: PersistentVolume
metadata:
  name: jenkins-pv
```

```
spec:
  capacity:
    storage: 10Gi
  volumeMode: Filesystem
  accessModes: [ReadWriteOnce]
  persistentVolumeReclaimPolicy: Retain
  storageClassName: jenkins-local
  local:
    path: /data/jenkins
  nodeAffinity:
    required:
      nodeSelectorTerms:
        - matchExpressions:
            - key: kubernetes.io/hostname
              operator: In
              values: [node01]
EOF
kubectl apply -f jenkins-pv.yaml
kubectl get pv
```

5. PersistentVolumeClaim

```
cat > jenkins-pvc.yaml <<'EOF'
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: jenkins-pvc
  namespace: jenkins
spec:
  accessModes: [ReadWriteOnce]
  storageClassName: jenkins-local
  volumeName: jenkins-pv
  resources:
    requests:
      storage: 10Gi
EOF
kubectl apply -f jenkins-pvc.yaml
kubectl get pv
kubectl get pvc -n jenkins
```

Expected: PV and PVC show Bound. A StorageClass object is not necessary for this explicitly bound static volume.

6. ServiceAccount and namespace RBAC

This grants only the permissions needed for a basic Jenkins Kubernetes-agent learning setup. Review and reduce permissions before production use.

```
cat > jenkins-rbac.yaml <<'EOF'
apiVersion: v1
kind: ServiceAccount
metadata:
  name: jenkins
  namespace: jenkins
---
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
```

```

name: jenkins-agent-manager
namespace: jenkins
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["get", "list", "watch", "create", "delete"]
- apiGroups: [""]
  resources: ["pods/log"]
  verbs: ["get", "list"]
- apiGroups: [""]
  resources: ["pods/exec"]
  verbs: ["create", "get"]
- apiGroups: [""]
  resources: ["events"]
  verbs: ["get", "list", "watch"]
---
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: jenkins-agent-manager
  namespace: jenkins
subjects:
- kind: ServiceAccount
  name: jenkins
  namespace: jenkins
roleRef:
  kind: Role
  name: jenkins-agent-manager
  apiGroup: rbac.authorization.k8s.io
EOF
kubectl apply -f jenkins-rbac.yaml

```

7. Headless Service for StatefulSet

```

cat > jenkins-headless.yaml <<'EOF'
apiVersion: v1
kind: Service
metadata:
  name: jenkins-headless
  namespace: jenkins
spec:
  clusterIP: None
  selector:
    app: jenkins
  ports:
    - name: http
      port: 8080
      targetPort: 8080
EOF
kubectl apply -f jenkins-headless.yaml

```

This enables a stable Pod DNS name such as `jenkins-0.jenkins-headless.jenkins.svc.cluster.local`. The separate NodePort Service provides browser access.

8. Deploy the Jenkins StatefulSet

```

cat > jenkins-statefulset.yaml <<'EOF'
apiVersion: apps/v1

```

```

kind: StatefulSet
metadata:
  name: jenkins
  namespace: jenkins
spec:
  serviceName: jenkins-headless
  replicas: 1
  selector:
    matchLabels:
      app: jenkins
  template:
    metadata:
      labels:
        app: jenkins
    spec:
      serviceAccountName: jenkins
      securityContext:
        runAsUser: 1000
        runAsGroup: 1000
        fsGroup: 1000
      containers:
        - name: jenkins
          image: jenkins/jenkins:lts-jdk21
          imagePullPolicy: IfNotPresent
          ports:
            - name: http
              containerPort: 8080
            - name: agent
              containerPort: 50000
          resources:
            requests:
              cpu: 500m
              memory: 1Gi
            limits:
              cpu: "2"
              memory: 3Gi
          volumeMounts:
            - name: jenkins-home
              mountPath: /var/jenkins_home
      startupProbe:
        httpGet:
          path: /login
          port: 8080
        periodSeconds: 10
        failureThreshold: 60
      readinessProbe:
        httpGet:
          path: /login
          port: 8080
        periodSeconds: 10
      volumes:
        - name: jenkins-home
          persistentVolumeClaim:
            claimName: jenkins-pvc
EOF
kubectl apply -f jenkins-statefulset.yaml
kubectl get pods -n jenkins -o wide -w

```

Expected: jenkins-0 reaches Running / 1/1 Ready on node01. For repeatable production deployments, pin a tested Jenkins image version or digest rather than a moving LTS tag.

9. NodePort Service for home-LAN access

```
cat > jenkins-service.yaml <<'EOF'
apiVersion: v1
kind: Service
metadata:
  name: jenkins
  namespace: jenkins
spec:
  type: NodePort
  selector:
    app: jenkins
  ports:
    - name: http
      port: 8080
      targetPort: 8080
      nodePort: 30080
EOF
kubectl apply -f jenkins-service.yaml
kubectl get svc -n jenkins
```

From a machine on your home LAN, visit <http://192.168.178.51:30080/>. Do not forward this unencrypted NodePort to the public internet. An Ingress/reverse proxy is optional for local learning, but useful later for HTTPS.

10. First login and plugin setup

```
kubectl exec -n jenkins jenkins-0 -- \
  cat /var/jenkins_home/secrets/initialAdminPassword
```

Open the Jenkins URL, paste the initial password, choose Install suggested plugins, create an administrator, and set the Jenkins URL to <http://192.168.178.51:30080/>. Do not store passwords or API tokens in Git.

11. Verify persistence and health

```
kubectl get all -n jenkins
kubectl get pv
kubectl get pvc -n jenkins
kubectl get pod jenkins-0 -n jenkins -o wide
kubectl exec -n jenkins jenkins-0 -- sh -c 'id; touch /var/jenkins_home/storage-test; ls -l /var/jenkins_home/storage-test'
```

Optional persistence test: after Jenkins is healthy, delete only its Pod, wait for the StatefulSet to recreate it, then confirm storage-test still exists. Do not delete the PVC or PV.

12. Troubleshooting

```
kubectl describe pod jenkins-0 -n jenkins
kubectl logs jenkins-0 -n jenkins
kubectl describe pvc jenkins-pvc -n jenkins
kubectl get events -n jenkins --sort-by=.lastTimestamp
kubectl get endpointslice -n jenkins
```

Pending Pod: check PV/PVC binding, node affinity, CPU/memory and taints. CrashLoopBackOff: inspect logs and `/data/jenkins` permissions. UI unavailable: check Service endpoints, NodePort 30080,

kube-proxy/CNI and host firewall. Plugin downloads failing: check DNS, outbound HTTPS and whether your network requires a forward proxy.

13. Security and next steps

Use HTTPS before exposing Jenkins outside a trusted LAN; limit who can reach NodePort; keep plugins updated; back up /data/jenkins; do not grant cluster-admin to Jenkins; and use separate short-lived Kubernetes agents for builds. On node02, image builds can use rootless BuildKit without installing a Docker daemon. The controller should not run regular build workloads.

14. Files and apply order

```
jenkins-pv.yaml  
jenkins-pvc.yaml  
jenkins-rbac.yaml  
jenkins-headless.yaml  
jenkins-statefulset.yaml  
jenkins-service.yaml
```

```
kubectl apply -f jenkins-pv.yaml  
kubectl apply -f jenkins-pvc.yaml  
kubectl apply -f jenkins-rbac.yaml  
kubectl apply -f jenkins-headless.yaml  
kubectl apply -f jenkins-statefulset.yaml  
kubectl apply -f jenkins-service.yaml
```

Reference: Kubernetes documentation for local PersistentVolumes, StatefulSets, Services and RBAC; Jenkins official container image documentation at github.com/jenkinsci/docker.